

Skills — Anthropic

Deep Agents ^{with} Skills

Claude Code

→ intro

→ summary of topic

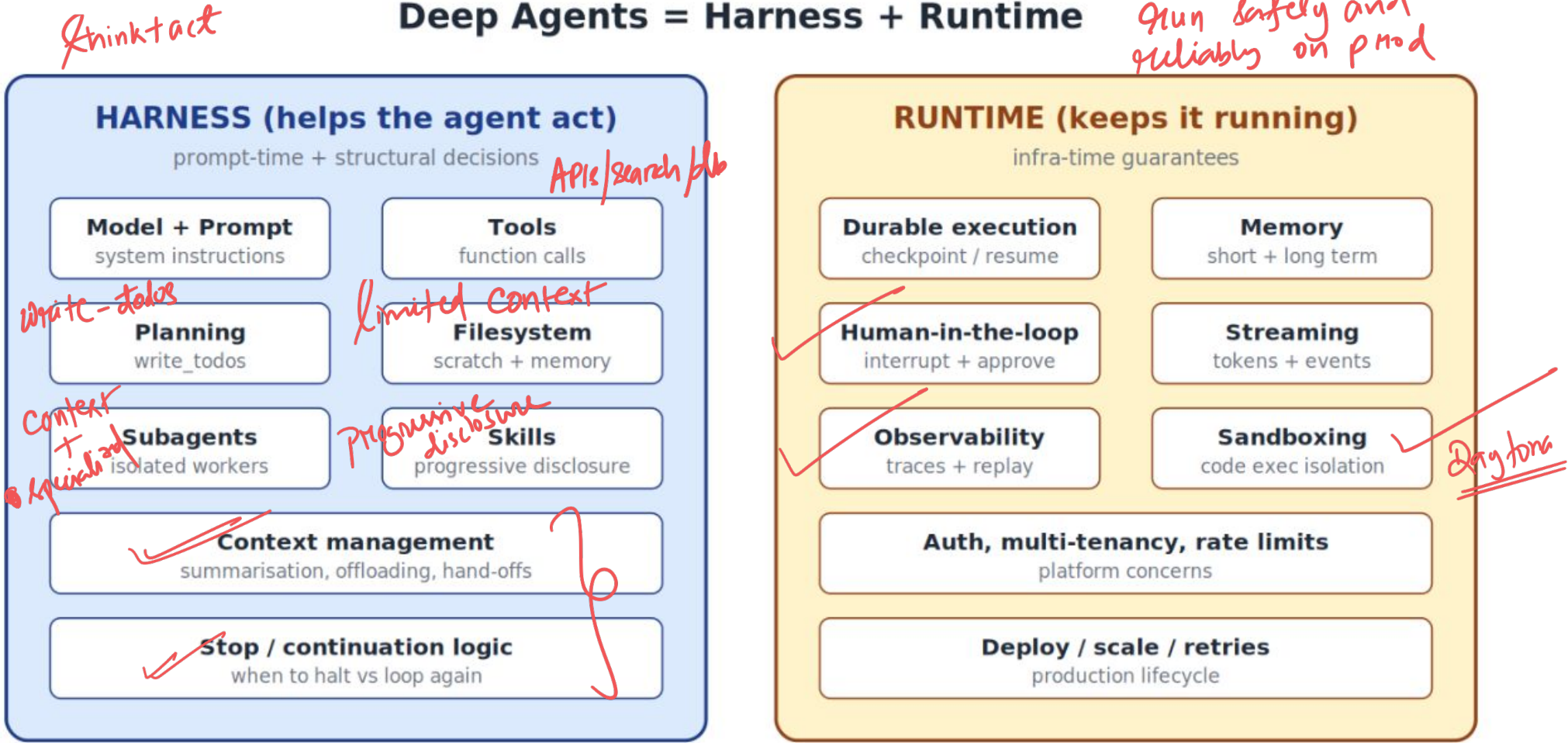
→ create im

→ create code

todos

Deep Agents = Harness + Runtime

run safely and reliably on prod

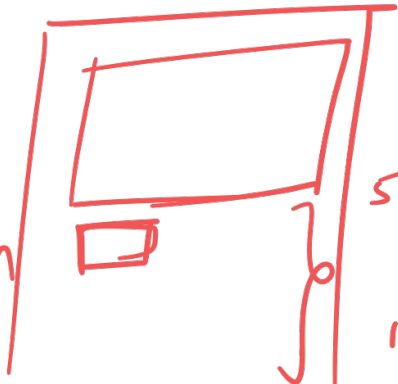


Specialize
Context

Supervisor



to Key



500

1000



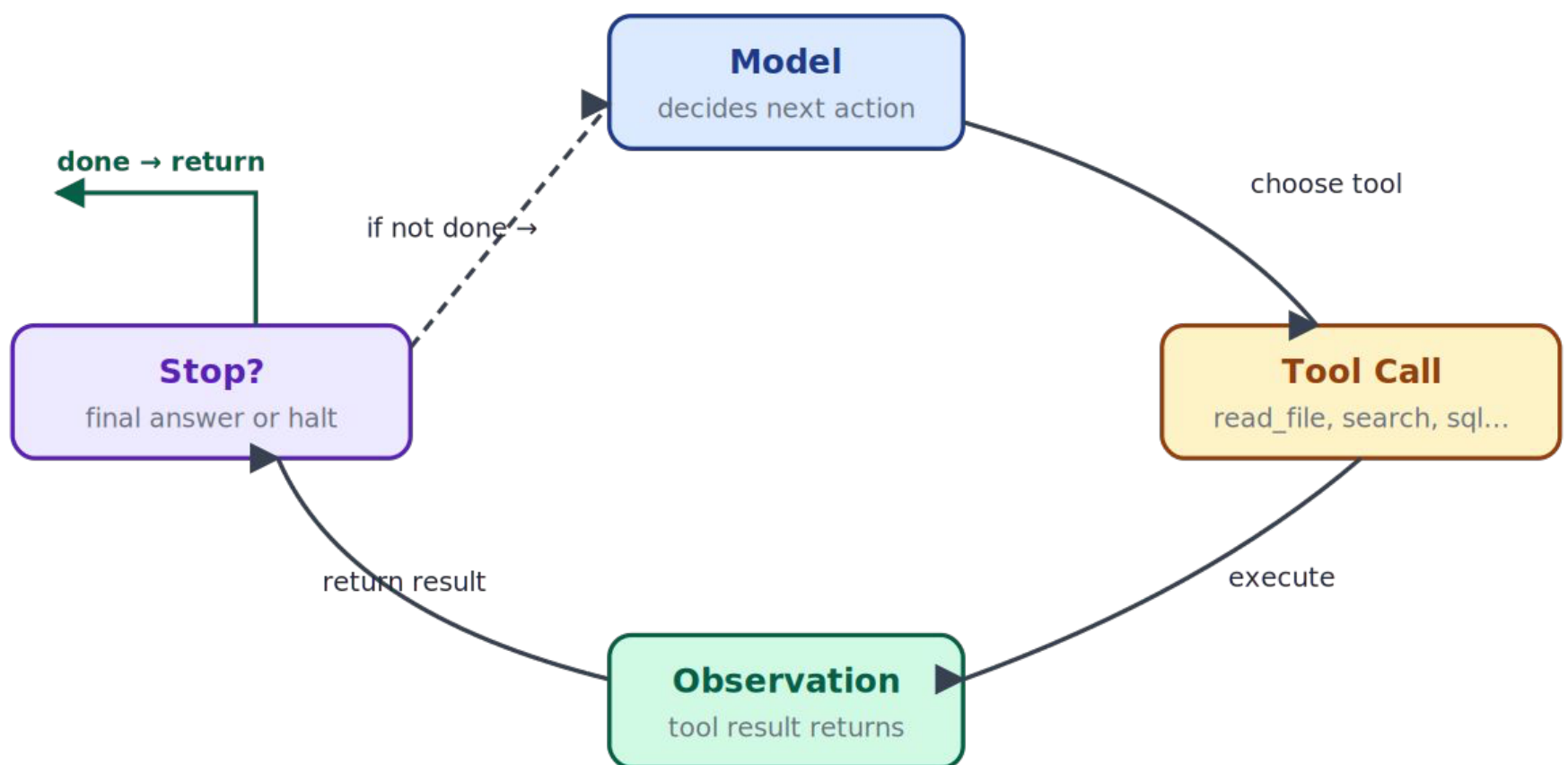
Subagent



1000



The Agent Loop

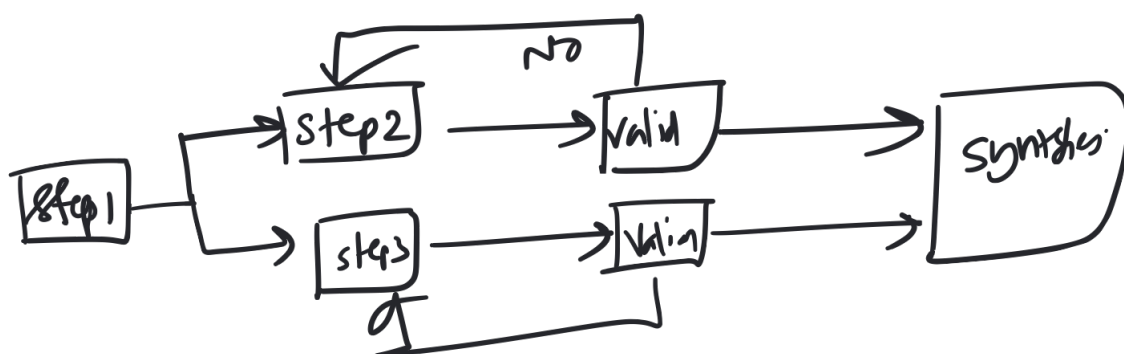


Primitives of Deep Agent

→ Todo

Langgraph

step1
step2/3
Synthesis



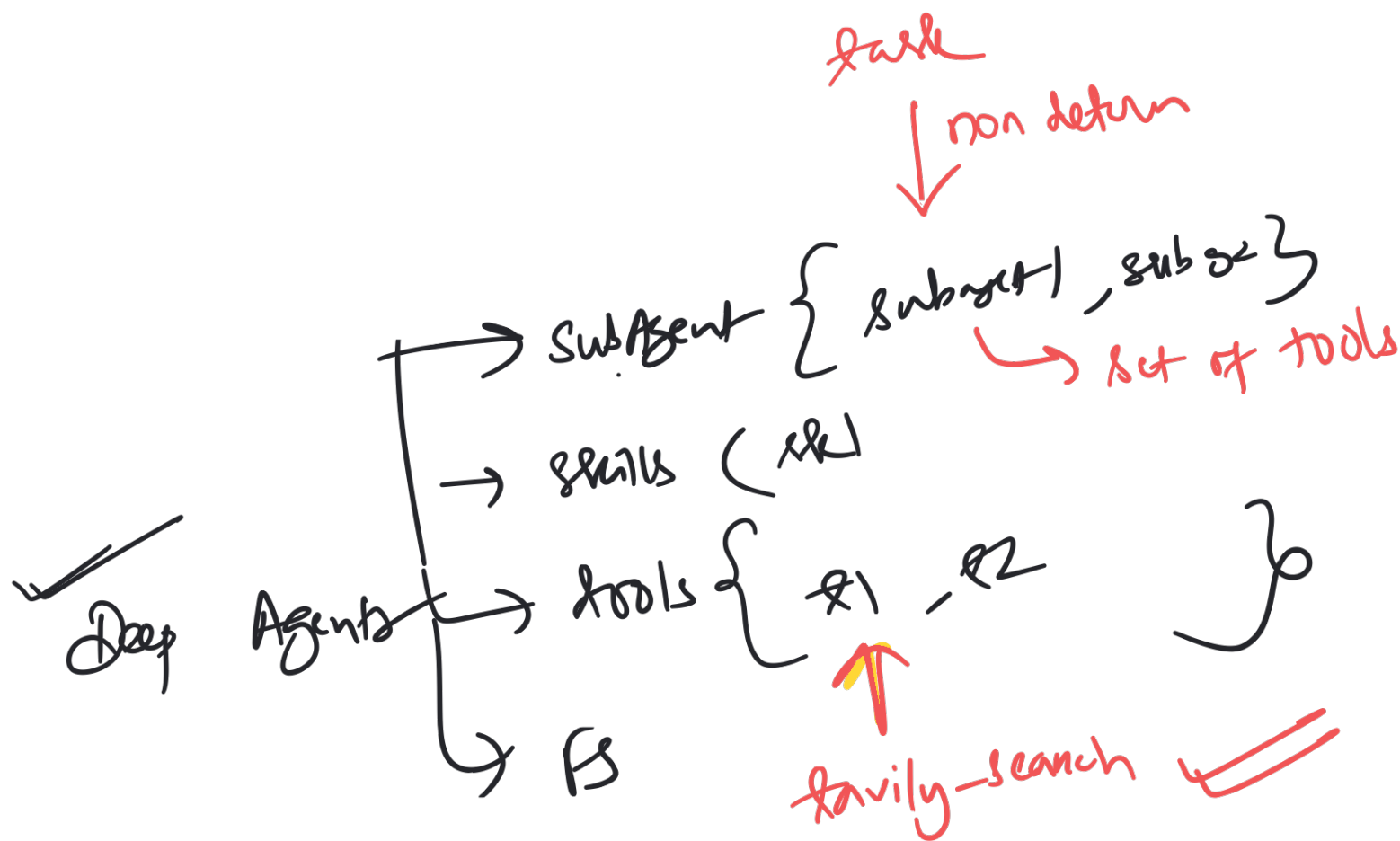
Workflow

steps are predetermined

Repo 1

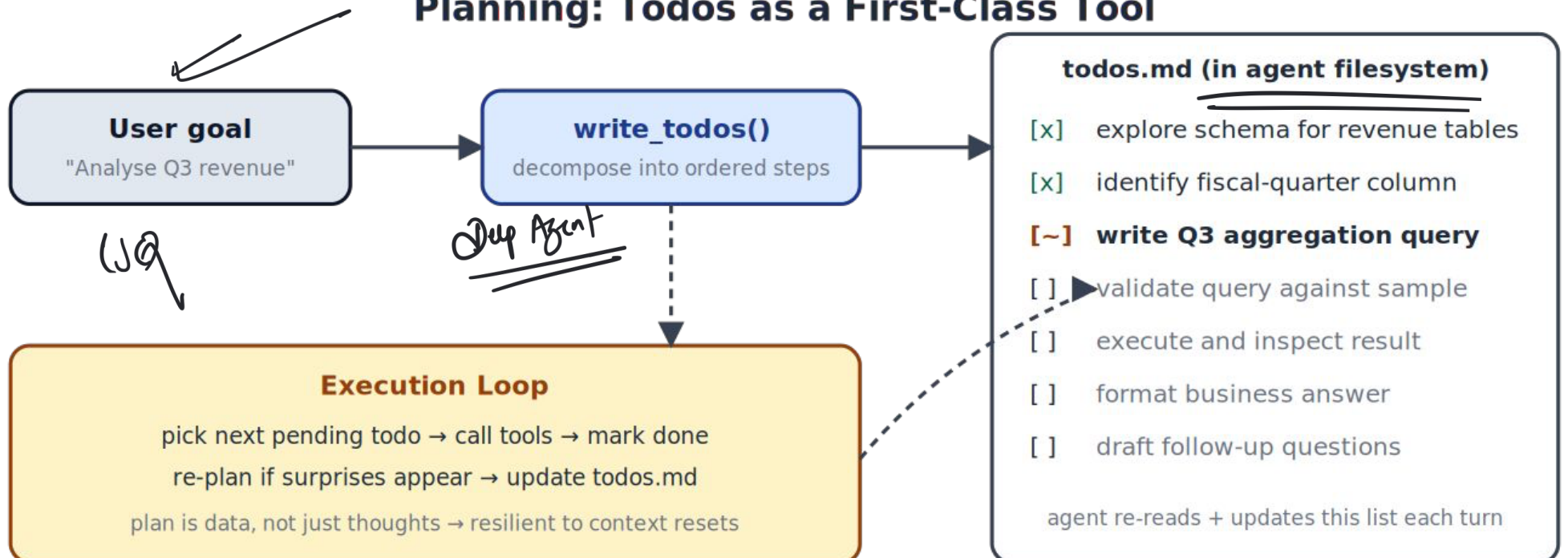
Repo 2

Agent



[] t2
[] t1
[] fs
[] subAgent1

Planning: Todos as a First-Class Tool



Context Bloat: Why Long-Horizon Agents Get Dumb

Without subagents

all tool calls share one context

main agent context window

user: do market analysis
tool_call search("vendor A")
+ 4KB raw HTML...
tool_call search("vendor B")
+ 6KB raw HTML...
tool_call search("pricing C")
+ 8KB raw HTML...
tool_call read_pdf(...)
+ 12KB extracted text...
tool_call read_pdf(...)
+ 14KB extracted text...

model now drowning in noise

- forgets goal, mis-cites, loops
- "context rot"

With subagents

noisy work isolated, only summary returns

main agent context

user: do market analysis
→ delegated to research-subagent
summary returned (≈ 800 tokens)
"3 vendors, A cheapest, B fastest..."

research-subagent (separate window)

tool_call search("vendor A")
tool_call search("vendor B")
tool_call search("pricing C")
tool_call read_pdf(...) ×2

noise stays here

- subagent returns synthesis only
- discarded after task completes

Agent

tool: description of tool

tool2: desc of tool2

System prompt

No Subagents

Main Agent

With Subagents

A1

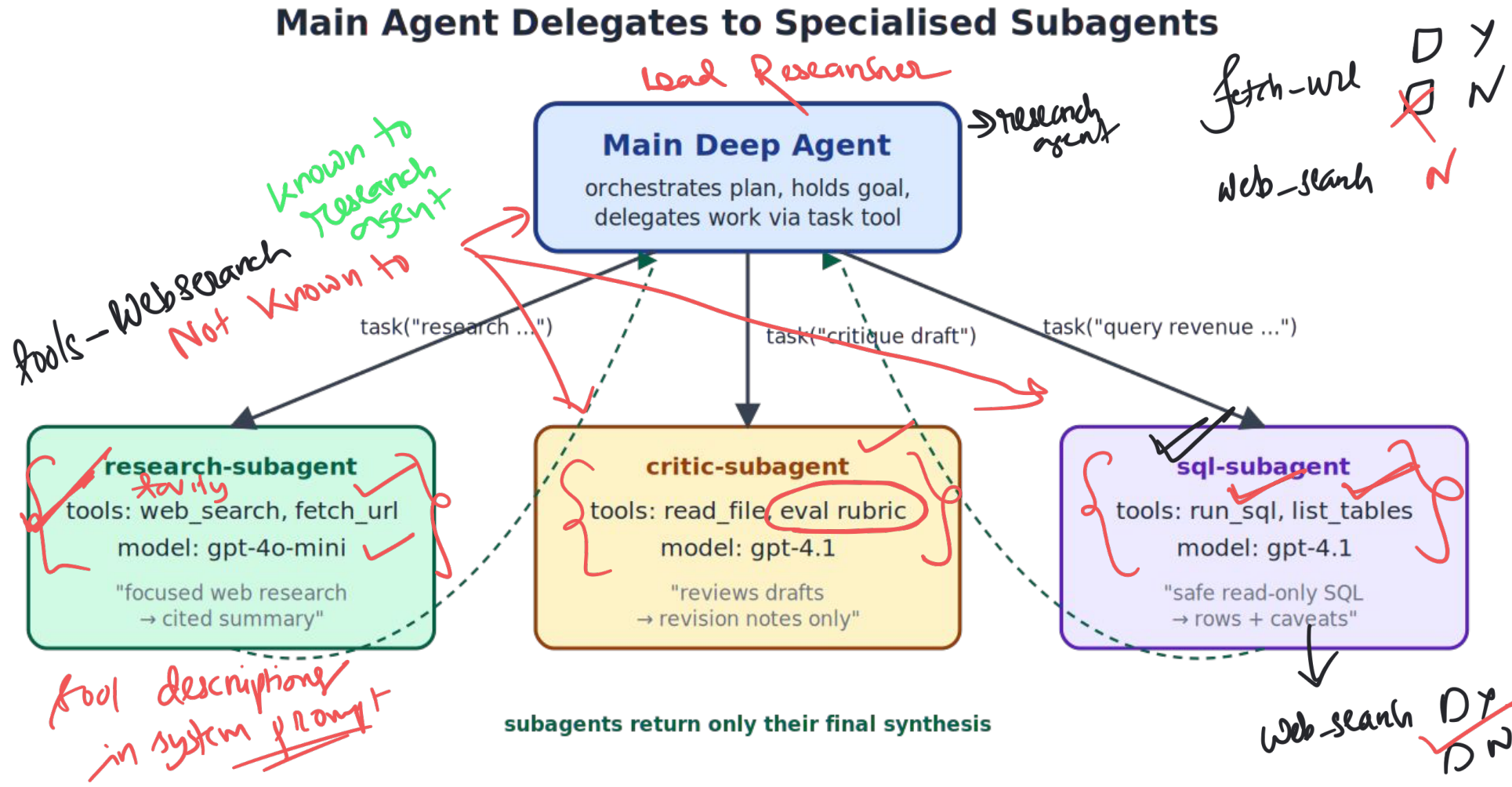
A2

A3

- web search, fetch url
- sql, execute sql
- eval metric

A1 desc
A2 desc
A3 desc

Main Agent Delegates to Specialised Subagents



No skills

Agent
System

Prompt -

- ✓ query explanation (SQL) → 100 words
- ✓ business information → 200 w
- ✓ query writing → 300 w

x 600

Skills

skill - ge - 100
skill - bf - 200
skill - qw - 300

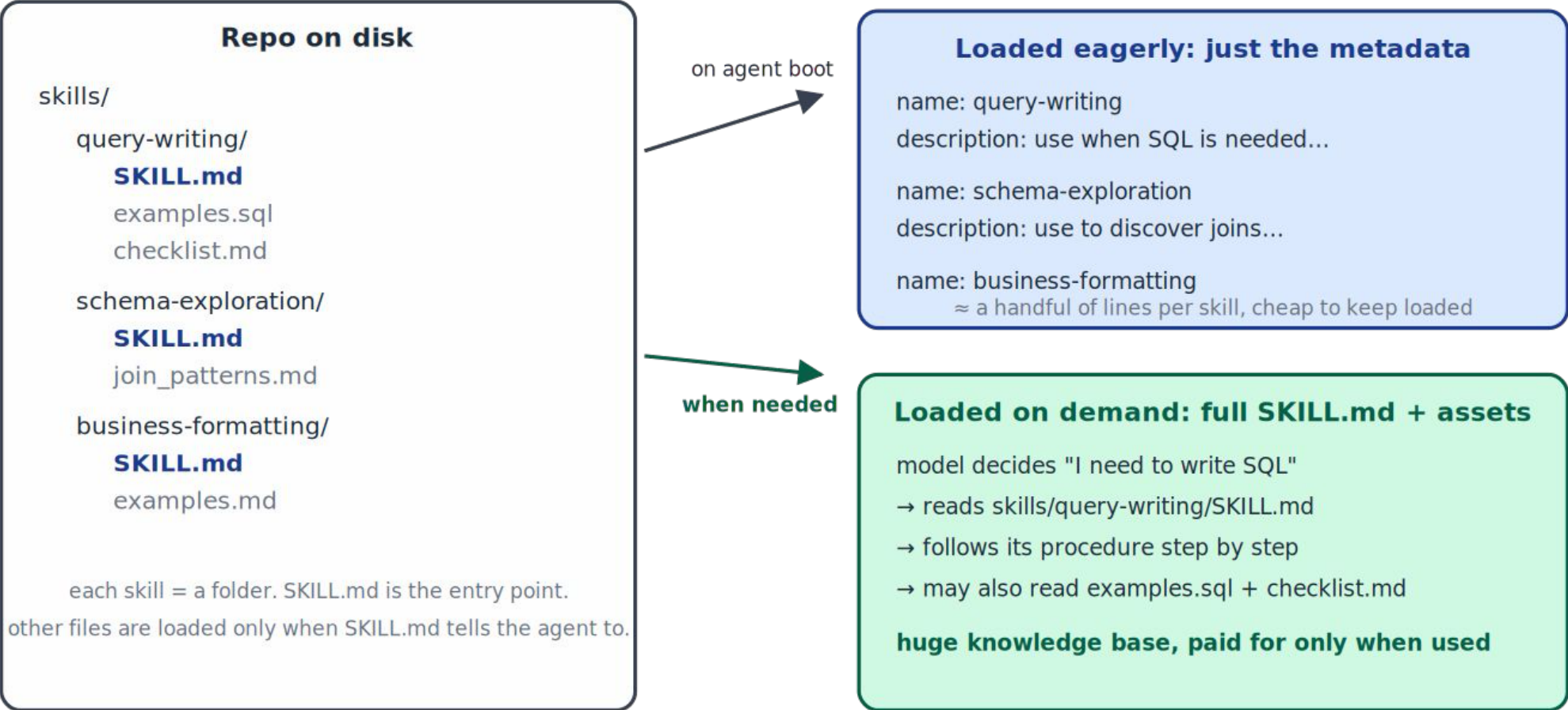
Agent
System prompt

ge skill - desc 10
bf skill - desc 10
qw skill - desc 10

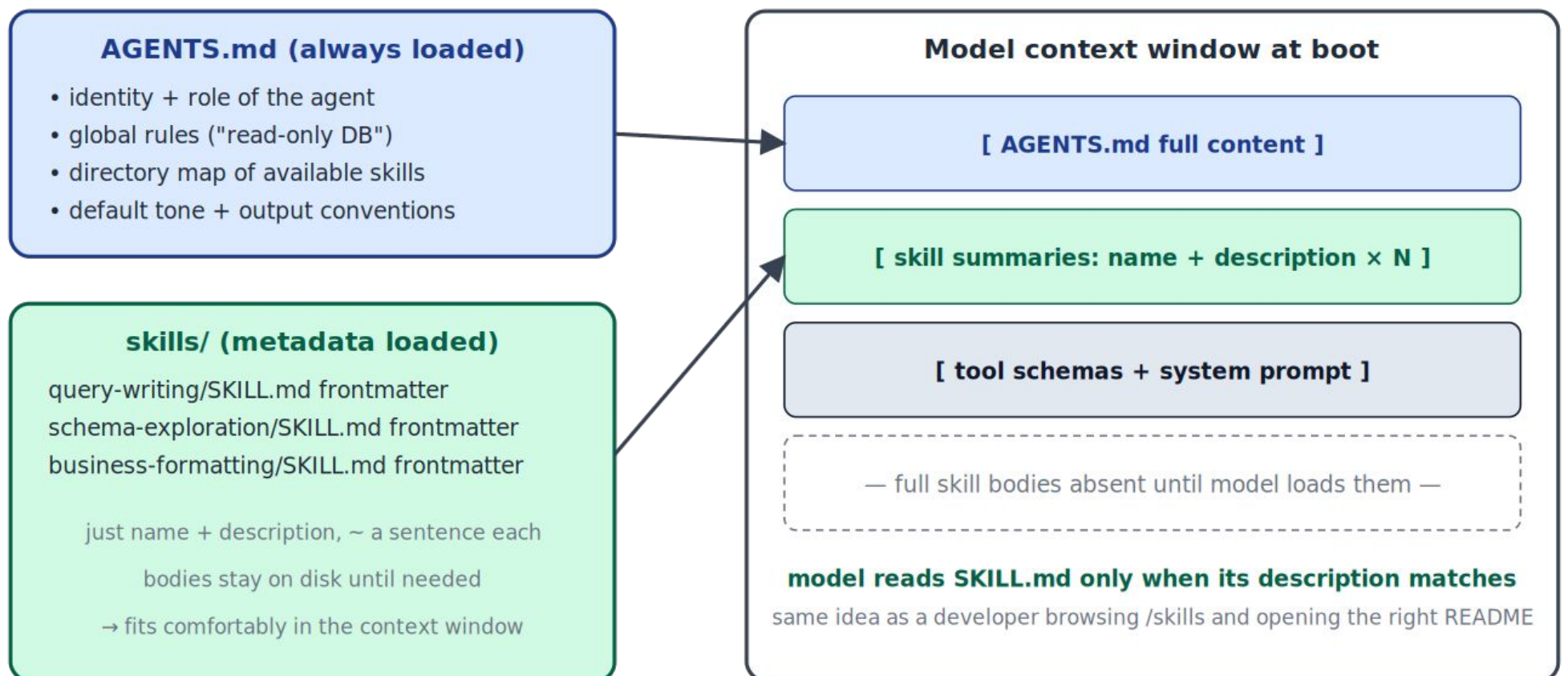
UQ → Agent → business format

30
+ 200
230

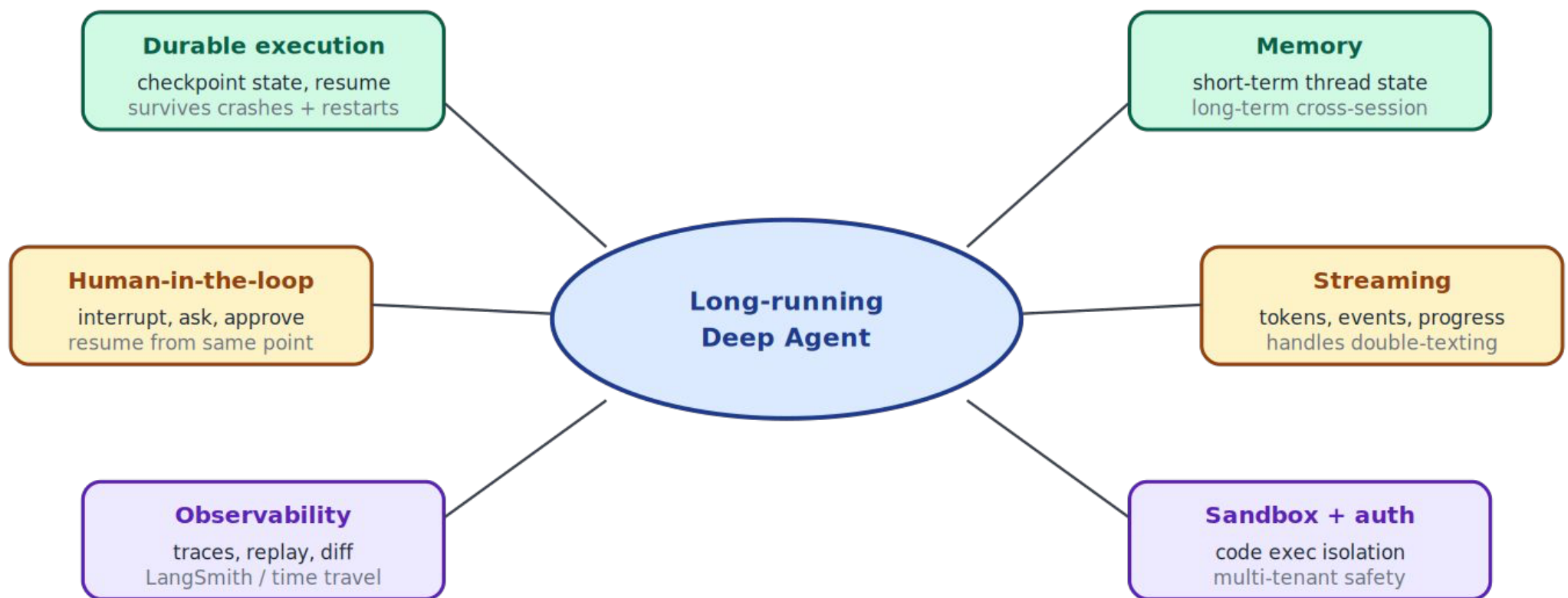
Skills: Progressive Disclosure via Filesystem

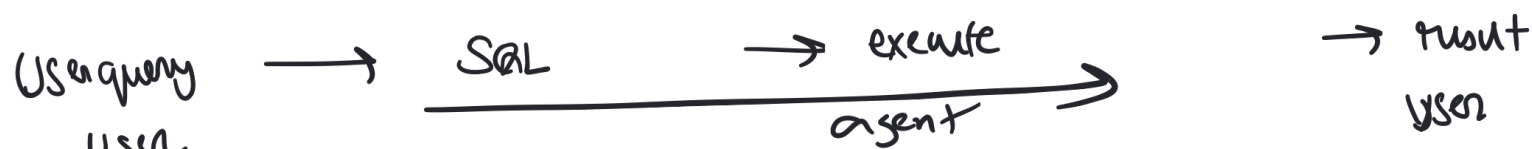


AGENTS.md + skills/ Loading Model



Runtime Capability Map





Text-to-SQL Deep Agent Architecture

